Reverse causality threatens half of CDC severe maternal morbidity indicators

More than half of CDC-defined severe maternal morbidity (SMM) events are not hospital-acquired or delivery-related outcomes — they reflect pre-existing conditions, antepartum complications, or unavoidable events that were present before any physician decision during delivery. This finding, from Fridman et al.'s landmark 2023 "performance SMM" analysis in California, fundamentally challenges any economics identification strategy that treats the full CDC SMM composite as an outcome of physician behavior. Jointcommission The clinical literature reveals that at least 9 of the 21 current CDC SMM indicators have high or very high antepartum onset probability, that obstetric physician assignment is systematically influenced by patient acuity through multiple documented mechanisms, and that emergency complications trigger non-random team formation. Taken together, these findings create serious reverse causality concerns: the patient's condition can cause the physician team composition, rather than the reverse.

More than half of CDC SMM is not caused by delivery care

The most powerful quantitative evidence comes from **Fridman, Korst, Reynen, Nicholas, Greene, Saeb, Troyan, and Gregory** (*Joint Commission Journal on Quality and Patient Safety*, March 2023), who developed a "performance SMM" (pSMM) measure by modifying the CDC composite. Yahoo! Their methodology excluded pre-existing conditions present on admission (using POA indicators), removed inconsistently documented outcomes, and risk-adjusted for conditions preceding hospitalization. ResearchGate +2 Using California hospital discharge data, the **pSMM rate was 0.44%, less than half the CDC SMM rate of 1.03%**. ScienceDirect The authors state explicitly: "Fewer than half of the observed cases of CDC SMM had the potential to be prevented through hospital practices." Jointcommission +2

This finding is corroborated by CMS itself. The CMS Severe Obstetric Complications eCQM (CMS1028) was redesigned to use present-on-admission codes, counting "only diagnoses not present on admission" as numerator events Qualityreportingcenter PubMed Central — an institutional acknowledgment that many SMM

diagnoses are present before delivery care begins. A Washington State study across 58 hospitals found that **23–39% of SMM diagnoses were coded as present on admission**, with higher POA rates at Level III/IV referral hospitals where complex patients concentrate. PubMed

ACOG's own Obstetric Care Consensus No. 2 (2016) states directly that "neither ICU admission nor transfusion of ≥4 units of blood should be used as quality metrics because some cases of morbidity reflect the underlying health of a woman or her pregnancy rather than the quality of care received." ACOG The leading SMM measurement researchers — Main, Fridman, Korst, Gregory, Howell, and Kilpatrick — have argued consistently that CDC SMM was designed for population surveillance, not facility or provider performance comparison. Jointcommission ScienceDirect

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Eclampsia, sickle cell crisis, and pulmonary edema violate the "outcome" assumption most severely

Among the 21 current CDC SMM indicators, the empirical evidence on antepartum onset is strongest for three conditions:

Eclampsia has the most robust timing data. Fishel Bartal and Sibai (*AJOG*, 2022) report that **up to 59–70% of eclamptic seizures occur during the antepartum period**. American Journal of Obstetrics … A multi-site analysis of 361 eclamptic women found 47% antepartum, 24% intrapartum, and 29% postpartum onset. PubMed Central StatPearls synthesizes the literature at 38–53% antepartum. NCBI Since delivery is the treatment for eclampsia (not its cause), a patient presenting with antepartum eclampsia triggers immediate mobilization of senior physicians, MFM specialists, anesthesiologists, and ICU teams — fundamentally altering team composition based on the patient's condition.

Sickle cell disease with crisis represents the clearest reverse causality case because SCD is a lifelong genetic condition that is always known before delivery. Per *Blood* (ASH, 2024), **50–70% of SCD pregnancies require at least one antepartum hospitalization**, and 30–40% require antepartum transfusion. American Society of Hematology The SMFM Consult Series #68 (*AJOG*, 2024) mandates management by "a multidisciplinary team that includes maternal-fetal medicine, hematology, genetics,

pain management, and social work." American Journal of Obstetrics … The physician team composition is entirely determined by the pre-existing diagnosis.

Pulmonary edema/acute heart failure shows 47% antepartum onset according to Sciscione et al. (*Obstetrics & Gynecology*, 2003), with pre-existing cardiac disease accounting for 25.5% of cases. Obstetrics & Gynecology Sibai et al. (*AJOG*, 1987) found 30% antepartum onset among preeclamptic-eclamptic patients with pulmonary edema, with 90% of those having pre-existing chronic hypertension. American Journal of Obstetrics … Patients with known cardiac disease are managed by dedicated "Pregnancy Heart Teams" Obstetrics & Gynecology per ACOG Practice Bulletin #212.

Disseminated intravascular coagulation is almost always secondary to an antepartum condition. Erez et al. (*AJOG*, 2015) document that the leading obstetric causes — placental abruption, preeclampsia/HELLP, retained stillbirth, acute fatty liver of pregnancy — are all antepartum. PubMed Sibai's HELLP data show **83% of patients already had DIC or suspected DIC at hospital admission** before any delivery decisions. PubMed Placenta accreta spectrum is similarly problematic: always diagnosed antepartum via ultrasound, Brigham and Women's Hospital it triggers assembly of massive multidisciplinary surgical teams Wiley Online Library including MFM specialists, gynecologic oncologists, urologists, interventional radiologists, and vascular surgeons University of Arizona College o… — all assembled weeks before delivery. Taylor & Francis Online OpenAnesthesia

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Obstetric physician assignment is systematically influenced by patient acuity

The assumption that attending and operating physicians are assigned via scheduling independent of patient risk is contradicted by the entire architecture of US maternal care. The **ACOG/SMFM Levels of Maternal Care framework** (Obstetric Care Consensus No. 9, 2019) establishes four tiers of facilities with escalating physician requirements, explicitly designed so higher-risk patients receive care from more specialized physicians. ASA Level III facilities require MFM specialists with inpatient privileges available at all times; Level IV centers require onsite medical and surgical subspecialty care.

The **SMFM Special Report** (*AJOG*, 2014) defines three escalating models of MFM involvement — consultation, co-management, and transfer of care — each triggered by patient acuity. Under co-

management, the MFM specialist handles the complex condition while a general OB handles routine care. Under transfer of care, the MFM "assumes complete responsibility" for all obstetric care including performing the delivery. A study at a teaching institution (*AJOG*, 2016) confirmed that both generalist OBs and MFM subspecialists rotate as laborists, meaning the attending physician's specialty varies and can be influenced by case complexity. PubMed Central American Journal of Obstetrics …

Validated obstetric triage systems further demonstrate acuity-based resource allocation. The **Maternal Fetal Triage Index** (MFTI) assigns patients to five acuity levels from "Stat" to "Scheduled," with ACOG explicitly stating that "women should be cared for according to triage acuity rather than by time of arrival" ACOG and evaluated "by a health care provider with skills and training appropriate to evaluate the issues identified during triage." ACOG Higher-acuity patients are systematically routed to more experienced providers. Between **45% and 55% of OB hospitalists act as "perinatologist extenders"** for MFM specialists, blurring the line between scheduled coverage and acuity-driven assignment. Contemporary OB/GYN

For cardiac patients specifically, the ACC Cardiovascular Disease in Women Committee Panel (*JACC: Advances*, 2024) recommends that mWHO category 3–4 patients receive care at Level IV centers with "on-site intensive care, cardio-obstetrics and maternal fetal medicine expertise, obstetric anesthesia, and cardiac surgery." Delivery location options range from standard L&D to cardiac operating rooms — all determined by patient risk, not physician schedules. PubMed Central

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Emergency complications trigger non-random team formation through multiple mechanisms

When emergencies occur during labor, team composition changes through at least four documented mechanisms that violate schedule-based assignment:

Category 1 cesarean declarations activate hospital-wide alert systems. The Rajagiri Hospital CODE-10 protocol specifies that "the primary surgeon/team leader is always an OBG consultant with 2–10 years of surgical expertise" PubMed Central — the most experienced available principle. Birmingham Women's NHS Foundation Trust mandates consultant notification for all emergency cesareans, with "failure to

inform the consultant" triggering incident reporting. Birthrights UK hospitals with off-site consultant call systems explicitly bring in senior physicians who would not otherwise be present, PubMed Central and these calls are triggered by patient urgency category.

Massive transfusion protocol activation summons multidisciplinary hemorrhage response teams including additional surgeons, anesthesiologists, hematologists, and blood bank personnel. The Indiana Maternal Hemorrhage Toolkit and Oregon Obstetric Hemorrhage Toolkit both specify hemorrhage response team compositions distinct from routine delivery teams. These are activated by hemorrhage severity — a patient condition, not a scheduling variable.

Obstetric rapid response teams such as Magee-Womens Hospital's "Condition O" (Gosman et al., *AJOG*, 2008) bring intensivists, attending anesthesiologists, and additional obstetricians to the bedside when clinical deterioration is detected. American Journal of Obstetrics … Some hospitals operate three separate code systems — obstetric emergency, cardiopulmonary compromise, and massive transfusion — each assembling different team compositions. Lippincott Williams & Wilkins ResearchGate Maternal early warning systems trigger escalation to MFM specialists, intensivists, and rapid response teams based on vital sign abnormalities. CEUfast

Perimortem cesarean and maternal cardiac arrest activate dedicated resuscitation teams per AHA guidelines, assembling obstetric, anesthesiology, neonatology, and critical care specialists. Baysidecpr +2 The presence of a **second/assistant surgeon** for cesarean is explicitly linked to anticipated complexity — insurance policies only reimburse assistant surgeon services when "medically necessary," and clinical examples document physicians being "called in" specifically because of complications encountered during surgery. MedibillMD

The failure-to-rescue framework explains why raw SMM conflates patient risk with care quality

The obstetric failure-to-rescue (FTR) literature provides the theoretical framework for understanding why treating all SMM as physician-caused is problematic. **Friedman, Ananth, Huang, D'Alton, and Wright** (*AJOG*, 2016) analyzed 50.4 million delivery hospitalizations PubMed and found that FTR — death

among women with severe morbidity — was the quality-sensitive metric, while complication occurrence was driven by patient factors. PubMed Central This mirrors the foundational surgical finding by **Silber et al.** (1992) and **Ghaferi et al.** (2009) that hospitals in the lowest and highest mortality quintiles had **virtually identical complication rates** but dramatically different rescue rates (12.5% vs. 21.4% FTR). PSNet

The preventability literature reinforces this distinction. Maryland hospital-based surveillance (Gyamfi-Bannerman et al., *JAMA Network Open*, 2022) found **~63% of SMM events were deemed preventable** through multidisciplinary review, but the most impactful intervention points were in the antepartum period and involved clinician-level factors — not delivery-room decisions alone. PubMed Central Econofact A New Zealand nationwide review (Lawton et al., *Acta Obstet Gynecol Scand*, 2019) found only **36% of SMM potentially preventable**, with preventability varying sharply by condition: sepsis at 56%, preeclampsia at 34%, major blood loss at 31%. Wiley Online Library The Post concept analysis (*Birth*, 2025) found FTR involves "multiple intersecting system-level breakdowns" rather than isolated provider errors. Gwu

Kern-Goldberger, Howell, Srinivas, and Levine (*AJOG MFM*, 2023) published a critical review of the SMM composite arguing it is a "heterogeneous composite mixing severity indicators AND interventions," that blood transfusion dominates the measure while often representing low-severity events, PubMed Central and that SMM was designed for surveillance rather than performance comparison. PubMed Central No published economics paper was found that has directly addressed the reverse causality concern specific to treating all SMM indicators as physician-caused outcomes, though **Currie and MacLeod** (*Journal of Labor Economics*, 2017) developed methods to decompose physician performance into diagnostic and procedural skill using cesarean data. University of Chicago Press

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A complete vulnerability ranking of the 21 current CDC SMM indicators

Based on the clinical evidence, the indicators can be ranked from least to most vulnerable to reverse causality. The ranking assesses three dimensions: (1) whether the condition can exist before delivery, (2) whether it triggers non-random physician team formation, and (3) whether it is primarily an outcome of delivery management.

Tier 1 — Least vulnerable (almost always post-delivery outcomes):

- **Severe anesthesia complications**: by definition occurs during/after anesthesia administration; team already in place
- **Heart failure during procedure/surgery**: ICD code designates intra-procedural onset
- **Amniotic fluid embolism**: 70% during labor, 19% during cesarean; unpredictable and unpreventable per SMFM
- **Conversion of cardiac rhythm**: resuscitative procedure performed after acute event; team already present
- **Temporary tracheostomy**: emergency airway procedure; reactive to acute events

Tier 2 — Low vulnerability (mostly acute delivery-related):

- **Acute myocardial infarction**: extremely rare; mostly acute events without antepartum warning
- **Cardiac arrest/ventricular fibrillation**: mostly acute; 38% hemorrhagic etiology per Mhyre et al. (*Anesthesiology*, 2014); PubMed Central triggers code team reactively
- **Shock**: mostly from postpartum hemorrhage; some antepartum causes from abruption or sepsis
- **ARDS**: mostly secondary to other acute peripartum events; some antepartum pneumonia/sepsis

Tier 3 — Moderate vulnerability (meaningful antepartum component):

- **Blood transfusion**: the most common SMM indicator, PubMed Central dominated by postpartum hemorrhage from uterine atony (~80% of PPH per ACOG), PubMed Central but antepartum hemorrhage (placenta previa, abruption) and pre-existing anemia create a meaningful reverse causality pathway. PubMed Central Main et al. found only **26% of transfusion-only SMM represented true severe morbidity**
- **Aneurysm**: can be known pre-existing condition; rare indicator overall
- **Puerperal cerebrovascular disorders**: mix of antepartum stroke (from preeclampsia) and postpartum events

- **Acute renal failure**: pre-existing CKD, preeclampsia-related renal insufficiency (diagnostic criterion for severe preeclampsia per ACOG), SCD nephropathy
- **Hysterectomy**: approximately **50% for placenta accreta spectrum** PubMed Central PubMed Central (Matsuo et al., *IJGO*, 2024: 53% elective), Wiley Online Library which involves planned multidisciplinary teams; PubMed the other half for uterine atony is a true delivery outcome ScienceDirect

Tier 4 — High vulnerability (frequently antepartum, triggers team changes):

- **Sepsis**: chorioamnionitis develops during labor before delivery decisions; triggers MFM, neonatology, infectious disease consultation
- **Air and thrombotic embolism**: antepartum VTE requires anticoagulation management and delivery planning by specialist teams
- **DIC**: leading causes are antepartum (abruption, preeclampsia/HELLP, retained stillbirth); Dovepress American Journal of Obstetrics … virtually always triggers multidisciplinary team activation Wiley Online Library
- **Mechanical ventilation**: critically ill patients may arrive already ventilated; pre-existing ventilation fundamentally determines team composition

Tier 5 — Very high vulnerability (predominantly pre-delivery, determines team):

- **Pulmonary edema/acute heart failure**: **47% antepartum onset**, 25.5% from pre-existing cardiac disease; PubMed Central ResearchGate Pregnancy Heart Teams assembled based on diagnosis UCLA Health
- **Eclampsia**: **50–70% antepartum onset**; Medscape triggers immediate senior physician mobilization; delivery is the treatment, not the cause TeachMeObGyn
- **Sickle cell disease with crisis**: lifelong pre-existing condition; Wiley Online Library **always** triggers multidisciplinary team assembly; American Journal of Obstetrics … 50–70% hospitalized antepartum;

American Society of Hematology RCOG the most unambiguous case of reverse causality in the entire SMM composite

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Practical implications for the identification strategy

The evidence suggests several concrete approaches for an economics study examining physician team gender composition and maternal outcomes. First, researchers should use the **Fridman pSMM framework** — excluding POA-coded conditions and pre-existing diagnoses ScienceDirect — rather than raw CDC SMM. This alone eliminates more than half of SMM events that cannot logically be attributed to delivery physician behavior. ScienceDirect Second, **sensitivity analyses excluding the Tier 4–5 indicators** (eclampsia, sickle cell, pulmonary edema, DIC, sepsis, thrombotic embolism, ventilation) would test whether results are driven by conditions that cause team formation rather than result from it. Third, **blood transfusion** deserves special attention: as the dominant component of the SMM composite, even its moderate vulnerability matters because antepartum hemorrhage from previa or abruption — conditions diagnosed before delivery — leads to both transfusion need and specific team assembly.

The strongest causal inference would come from restricting the outcome to Tier 1–2 indicators (severe anesthesia complications, amniotic fluid embolism, cardiac arrest, shock, ARDS) that are almost exclusively acute consequences of delivery management, though these are rare enough to create statistical power concerns. The CMS approach of using POA indicators to define hospital-acquired complications offers a data-driven solution if discharge records include POA flags. Qualityreportingcenter PubMed Hospitals using laborist models with truly schedule-based assignment would provide the cleanest natural experiment, PubMed Central +2 but even in these settings, the SMFM framework for escalating MFM involvement based on acuity means the "operating physician" recorded in claims data may not be the scheduled laborist when complications are anticipated or detected. ScienceDirect